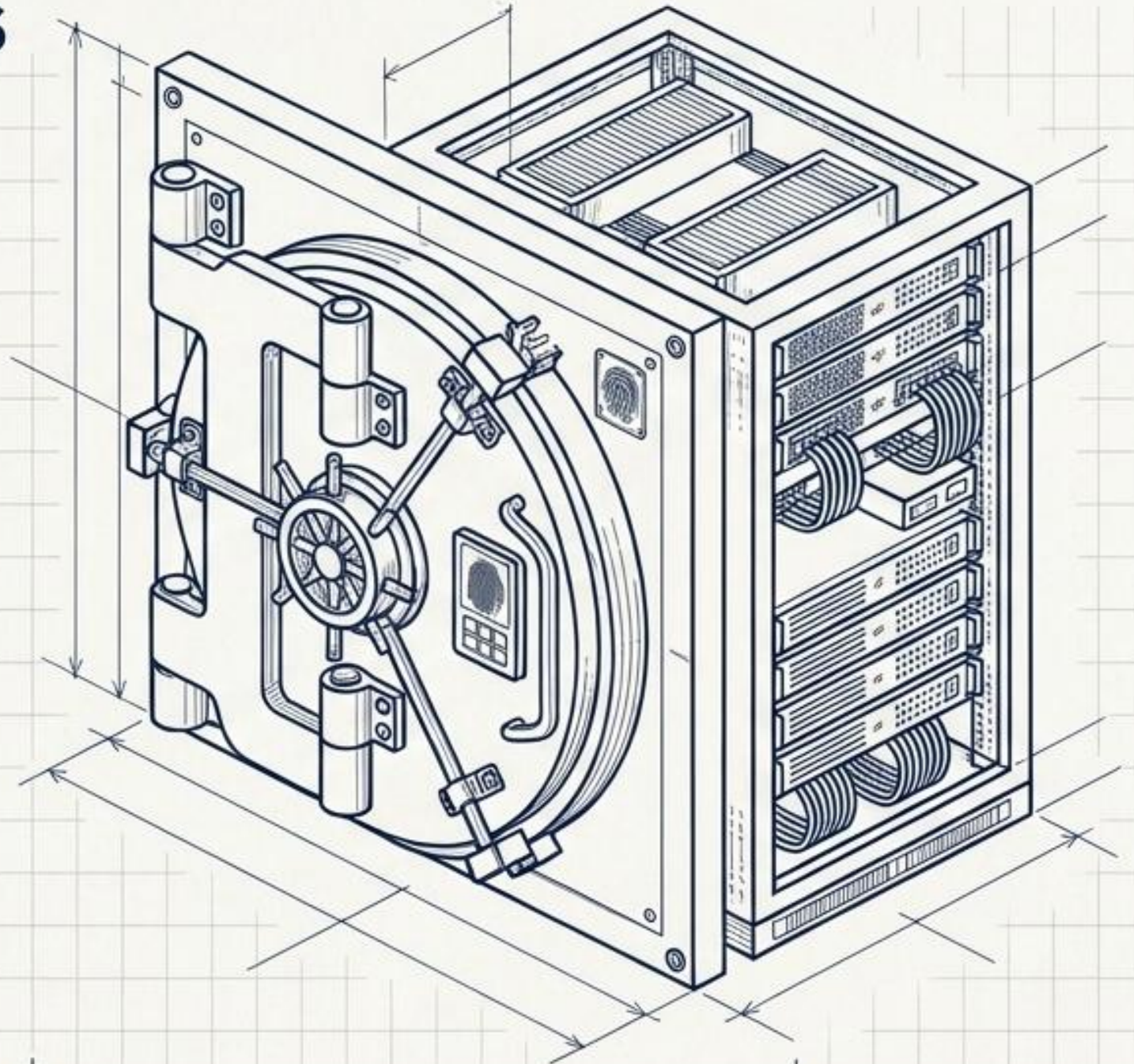


Computer System Security

Building the Digital Fortress

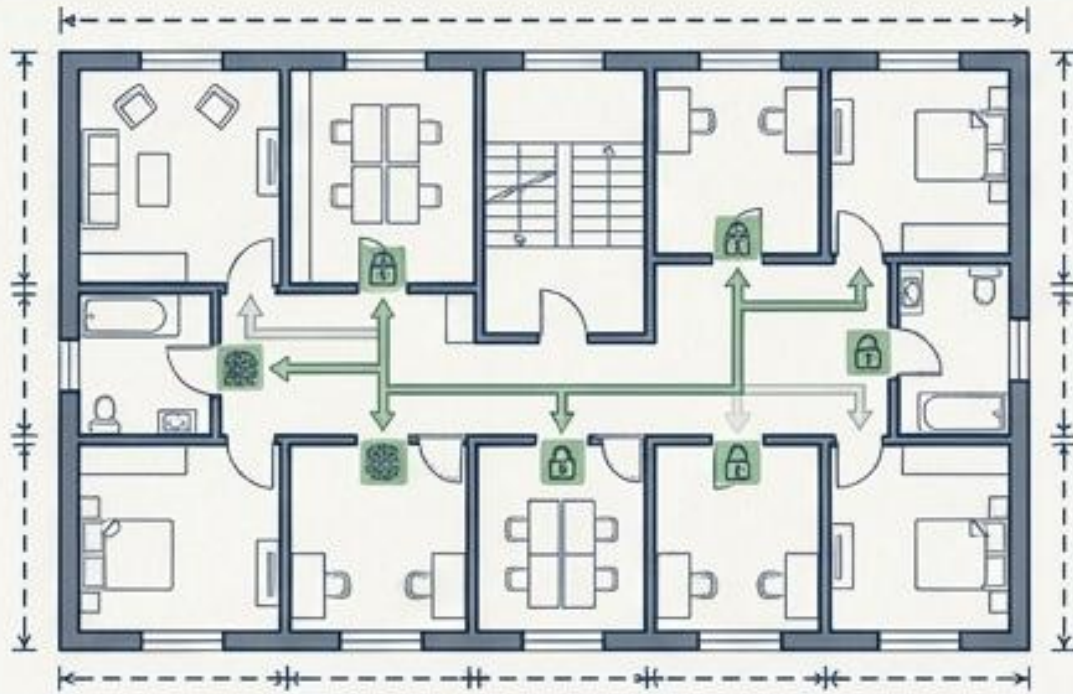


Prepared by: Hossam Nour, Ahmed El-Desouky, Anas Saber

Course: Security

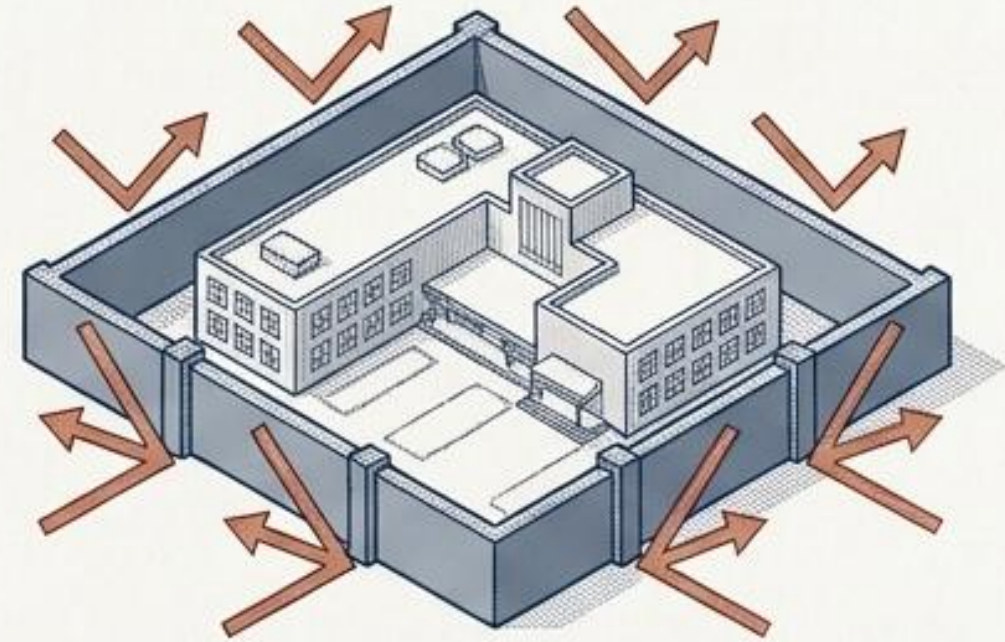
Date: 2026

Defining the Perimeter



Protection (Internal Focus)

- **Target:** Unauthorized internal access. Ensures everyone stays in their lane.
- **Mechanism:** Access Control Lists (ACLs).
- **Example:** File Permissions (Read, Write, Execute).



Security (External Focus)

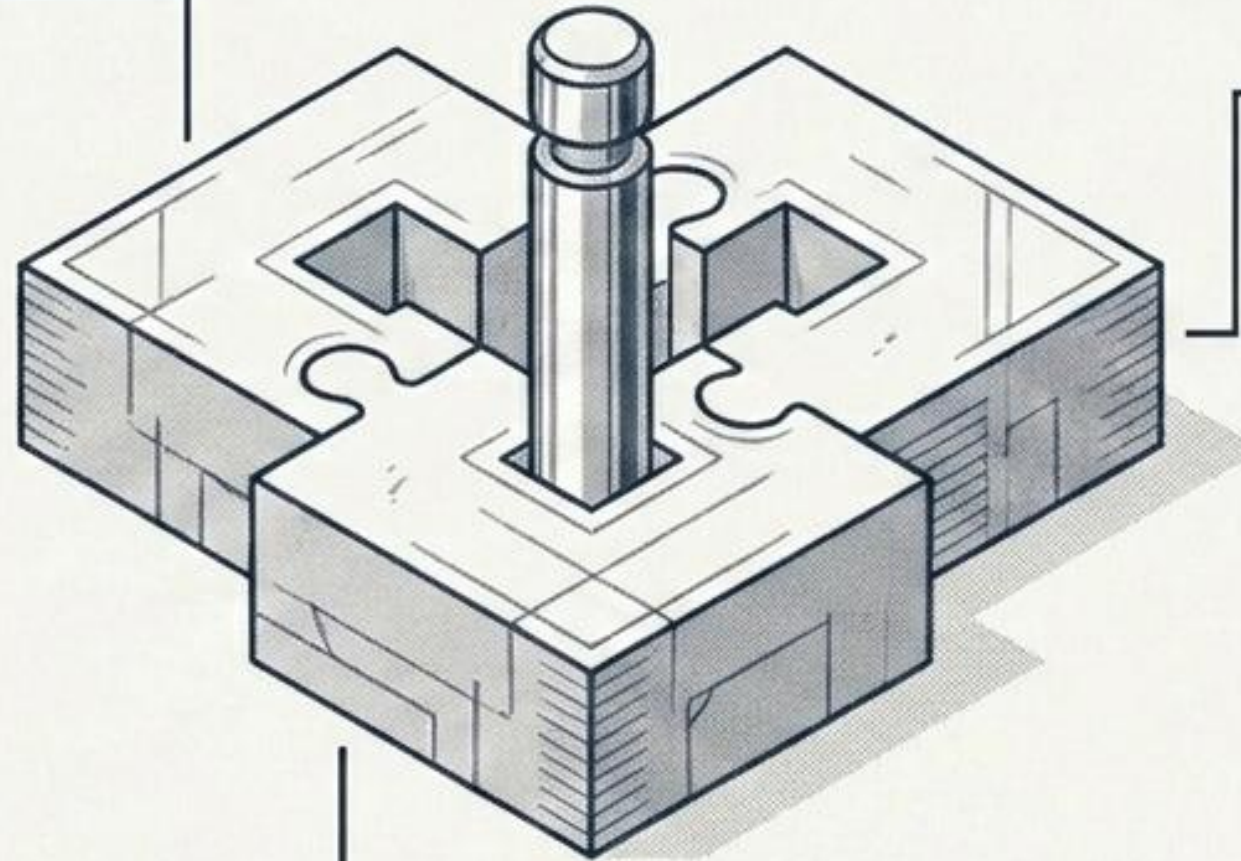
- **Target:** External threats and malicious intruders. Keeps the bad guys out.
- **Mechanism:** Authentication & Threat Prevention.
- **Examples:** Passwords, Firewalls, and Encryption.

**Protection enforces rules inside the system;
Security builds a wall around it.**

The Core Nucleus

Confidentiality

Only authorized users
can see the data.



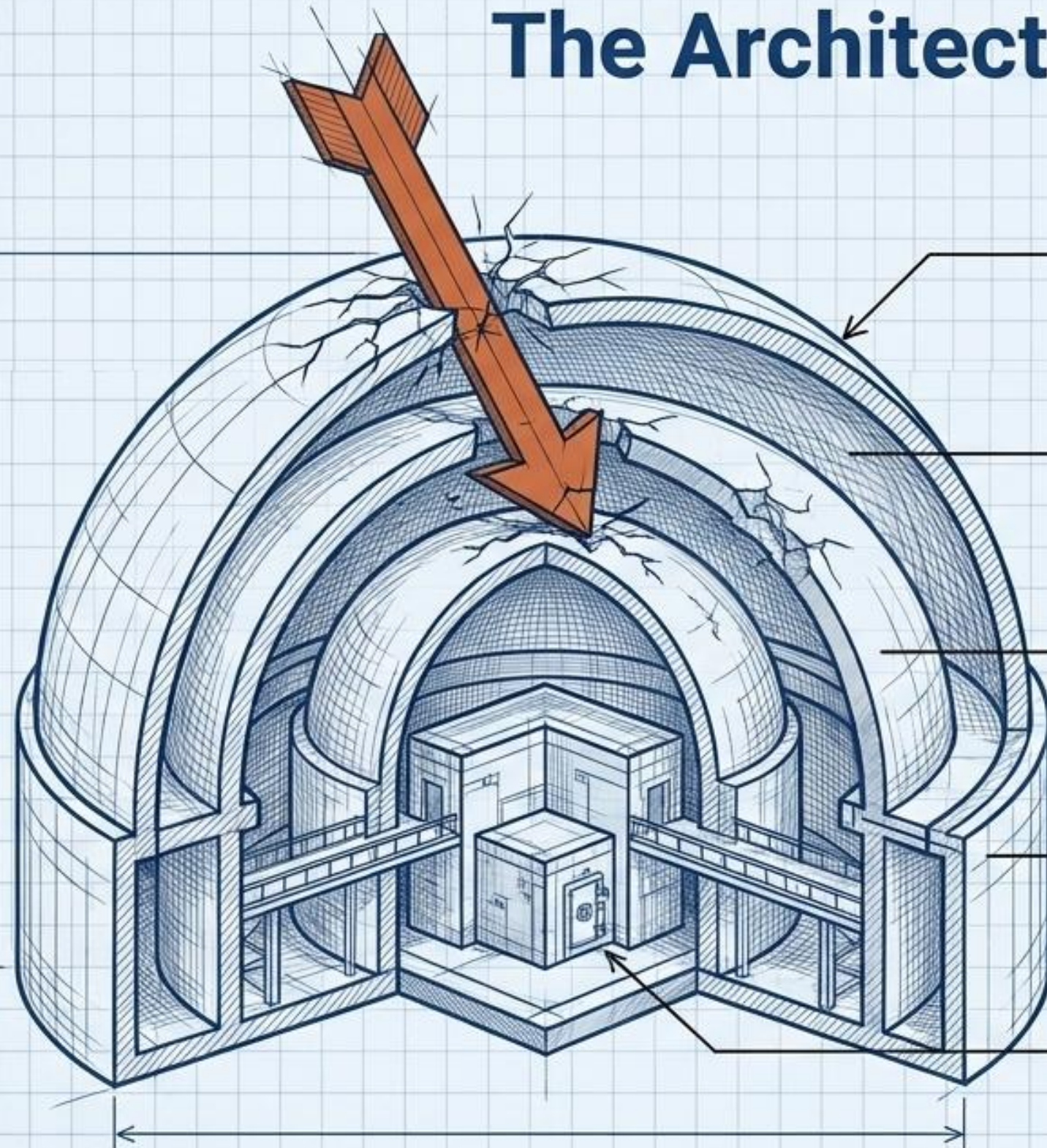
Integrity

Protect data from
unauthorized changes.

Availability

Services and resources work when needed.

The Architecture of Defense



Physical Level: Securing tangible devices and buildings (gates, cameras).

Human Level: User awareness and behavioral security against social engineering.

Network Level: Securing data during transmission between systems (firewalls, encryption).

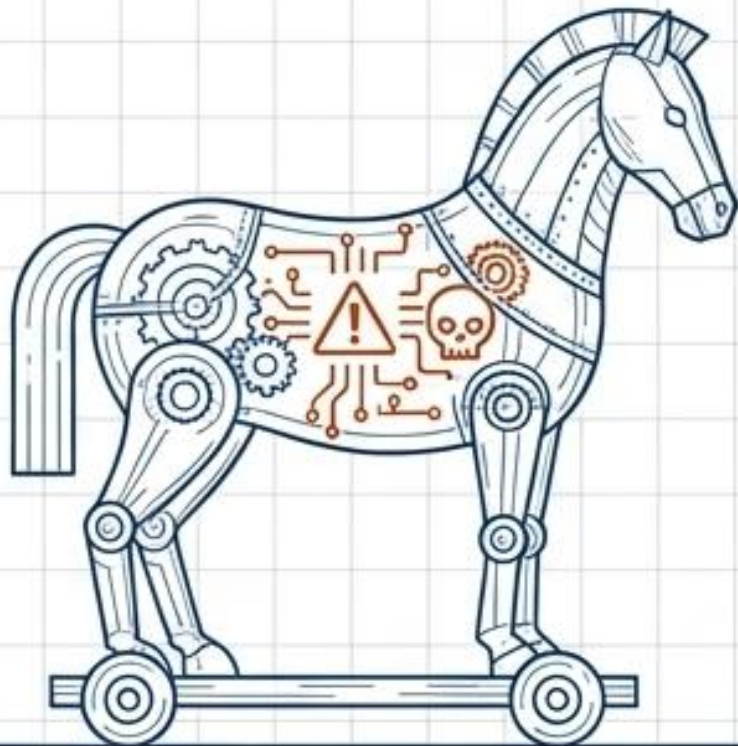
OS Level: Protecting the environment where programs run (updates, access tools).

Core: Data / Resources.

Vectors of Compromise



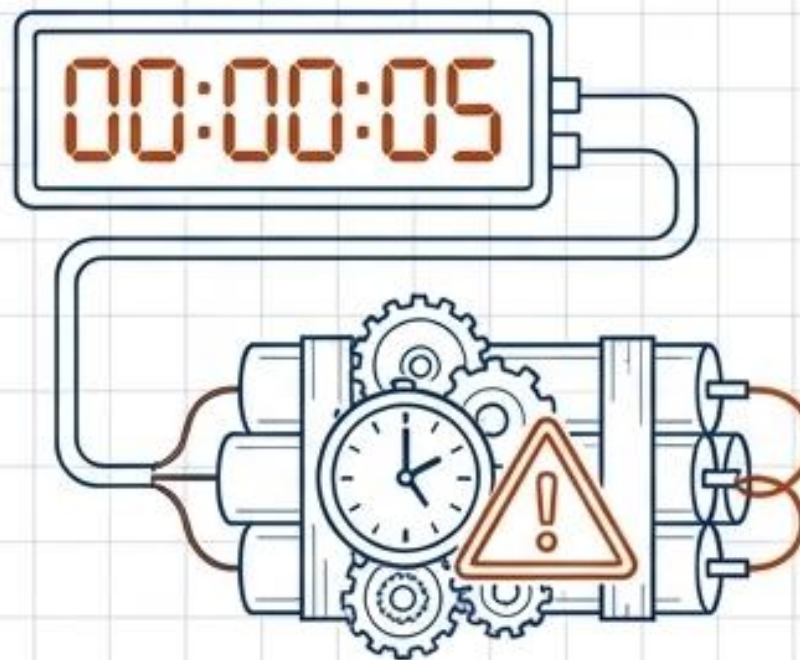
Trojan Horse



Type: Disguised malware.

Description: Appears as a useful or legitimate program but contains malicious code to open system vulnerabilities upon execution.

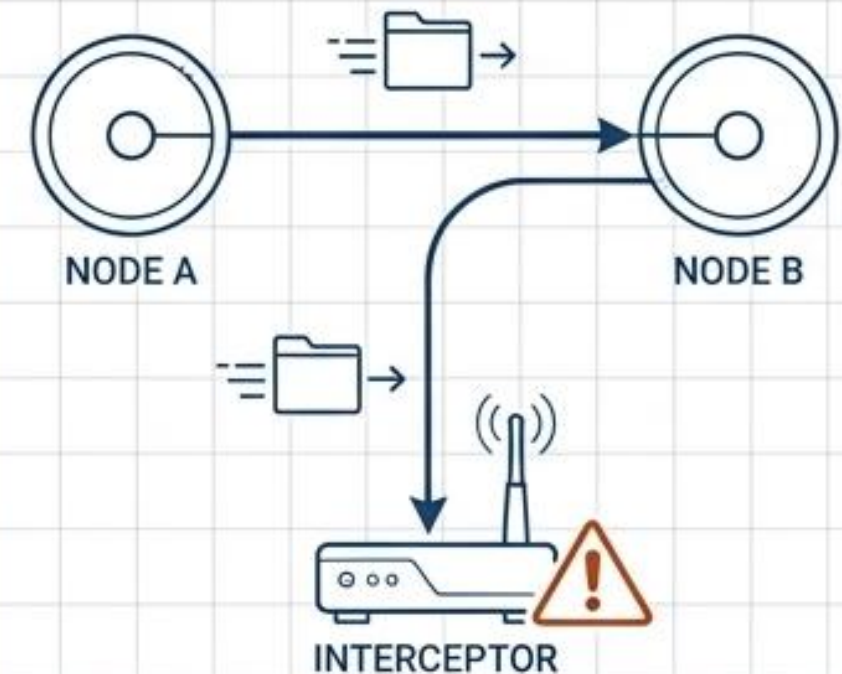
Logic Bomb



Type: Time/Event-triggered attack.

Description: Malicious code that lies dormant within a system until triggered by a specific event or date to destroy data or crash the system.

Interception



Type: Network and data breach.

Description: Eavesdropping on data during its transmission across a network, allowing unauthorized entities to steal or read it.



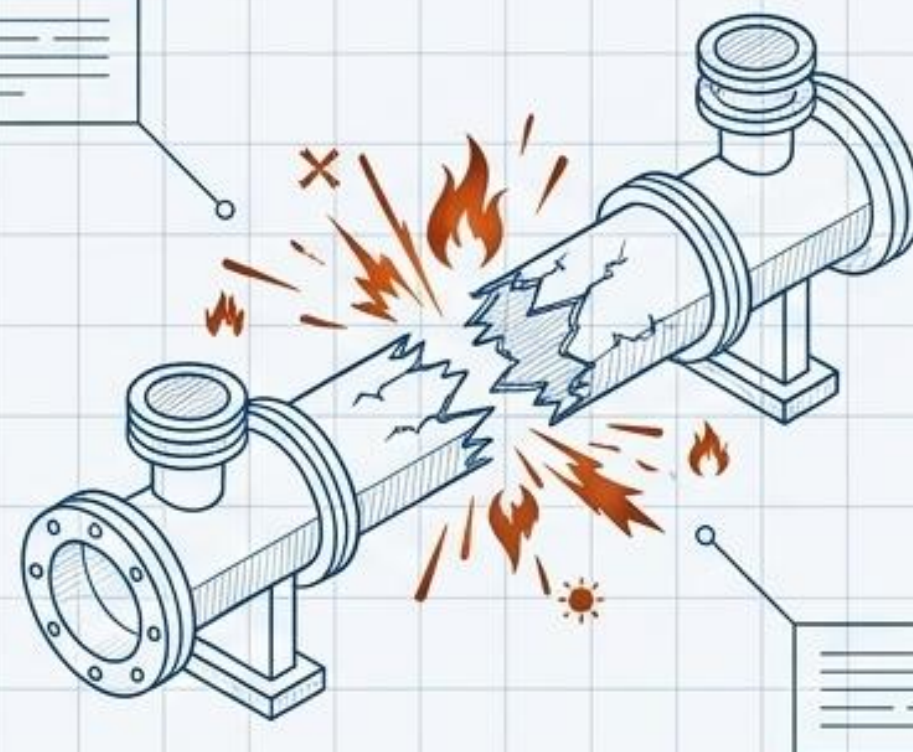
The Battlefield: When Defenses Fail

WannaCry Ransomware



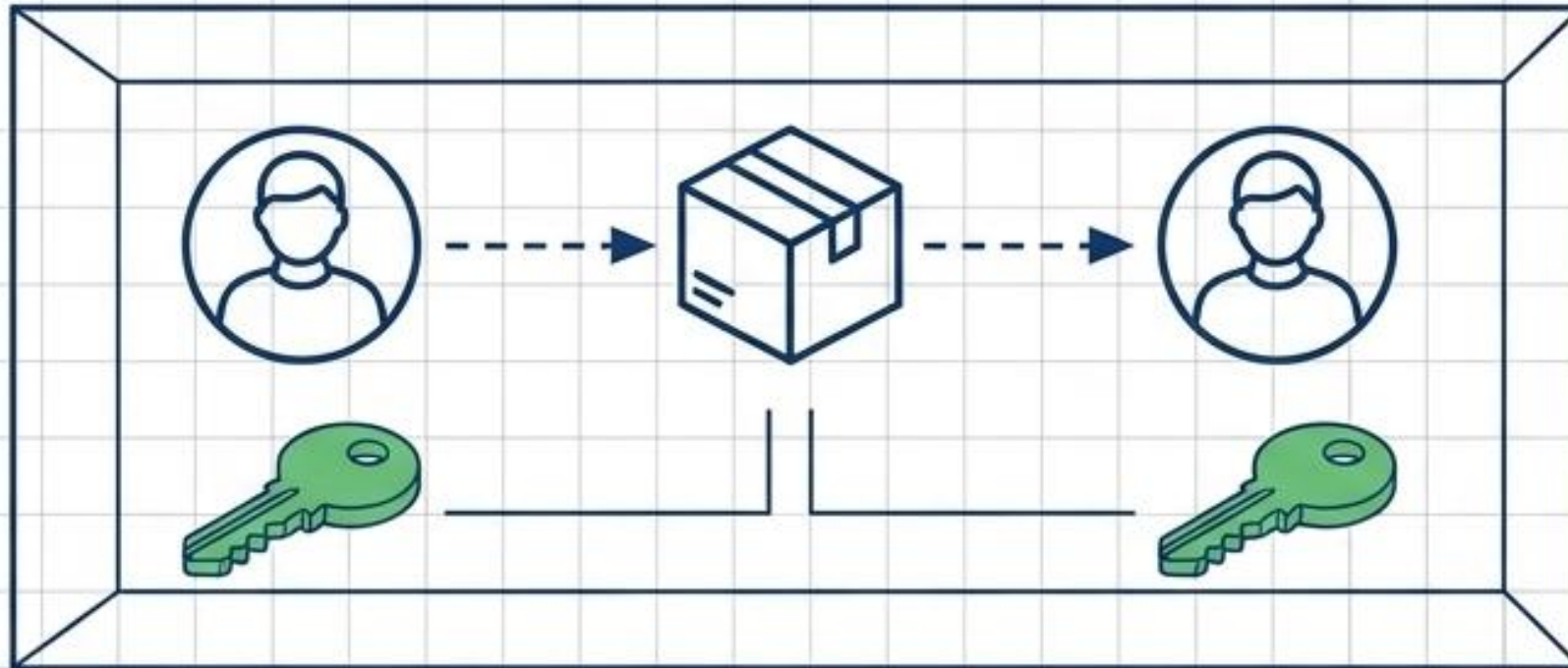
- **Event:** A globally spread ransomware attack.
- **Impact:** Encrypted user and institutional data (including hospitals) and demanded financial payout for recovery.
- **Vulnerability Exploited:** Bypassed the OS Level and Network Level.

Stuxnet Worm



- **Event:** Highly complex computer worm designed for physical sabotage.
- **Impact:** Targeted industrial control systems (SCADA), causing actual physical damage to centrifuges.
- **Vulnerability Exploited:** Bypassed the Human and Physical levels via a compromised USB drive to reach the OS.

The Shield: Securing the Data

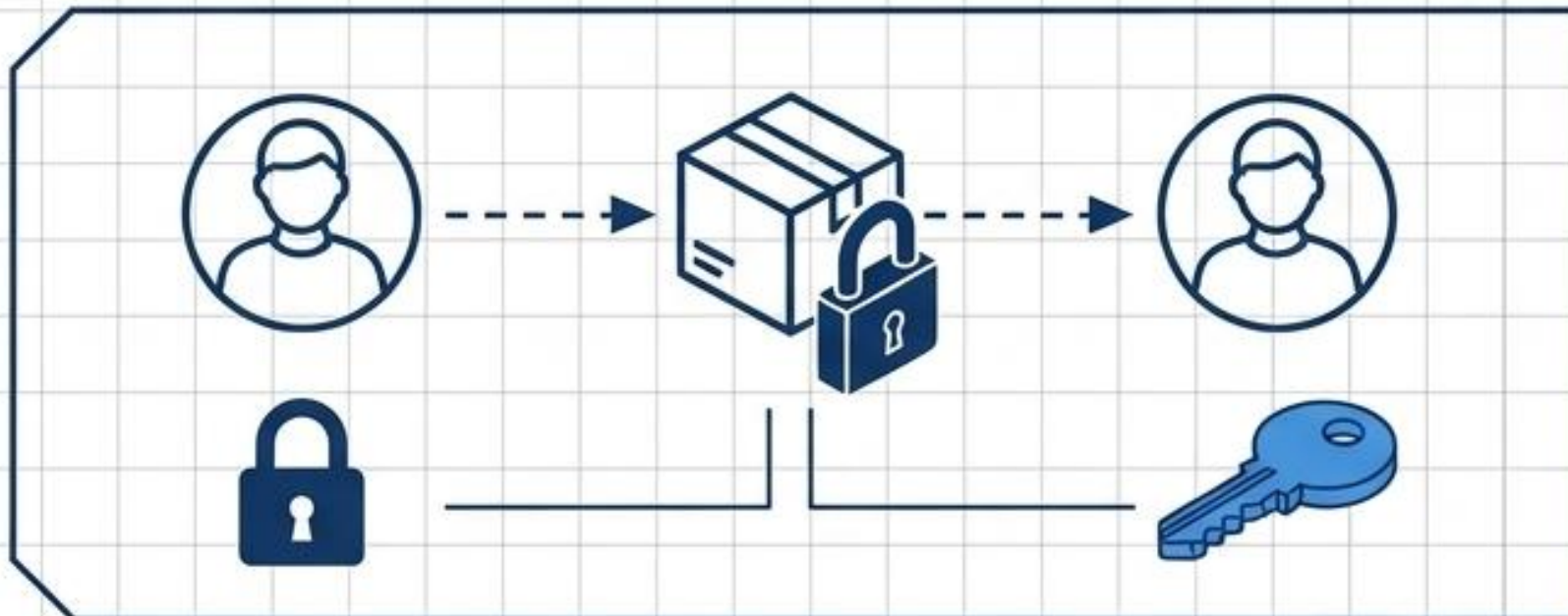


AES (Advanced Encryption Standard)

Structure: Symmetric Encryption.

Mechanism: Uses 1 Key for both encryption and decryption.

Performance: Very fast, making it ideal for large volumes of data.



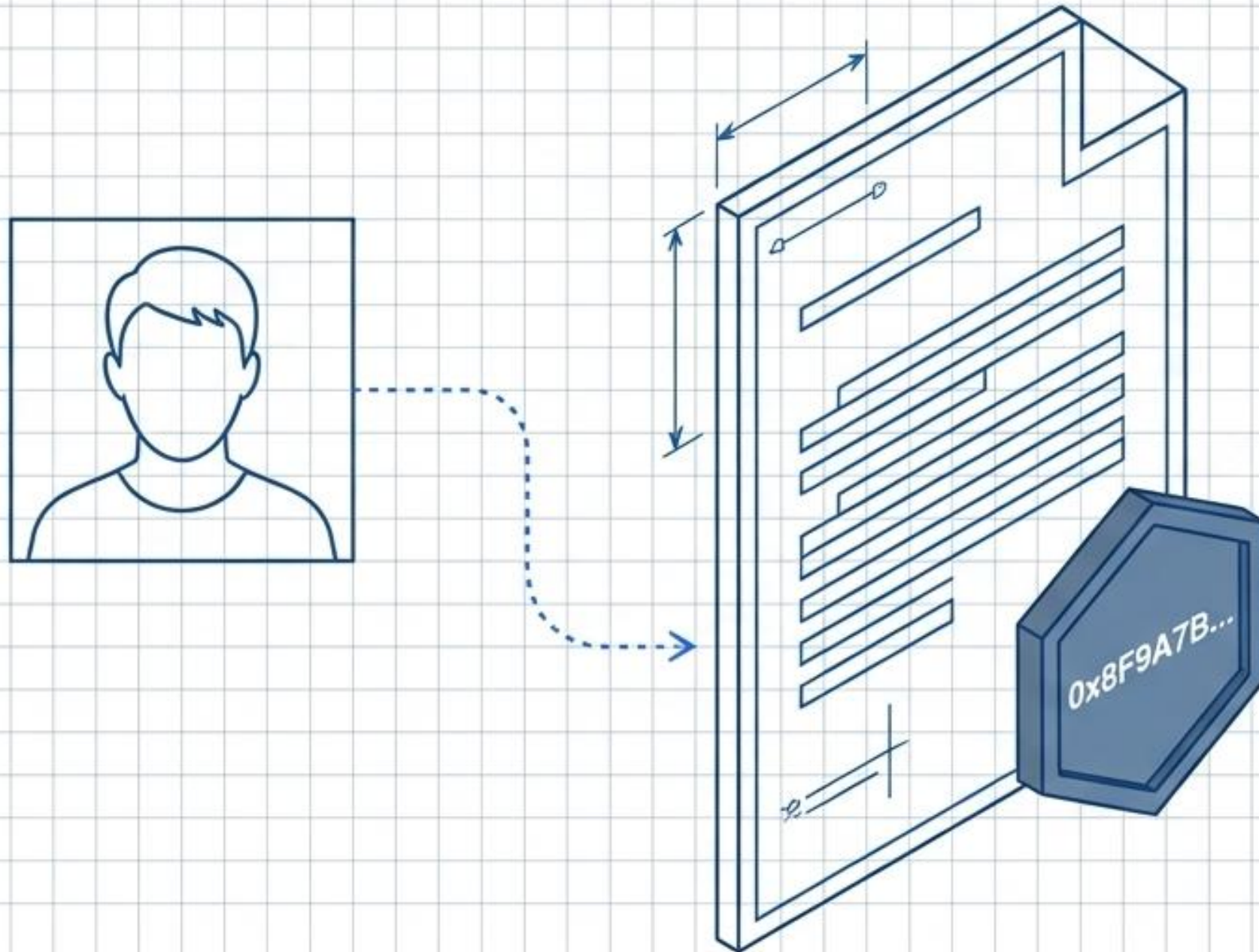
RSA (Rivest-Shamir-Adleman)

Structure: Asymmetric Encryption.

Mechanism: Uses 2 Keys – a public key for encryption, and a private key for decryption.

Performance: Highly secure, primarily used for the safe exchange of keys and sensitive data.

Digital Identity and Integrity



- **Beyond Encryption**

It is not just about protecting the data, but guaranteeing its source.

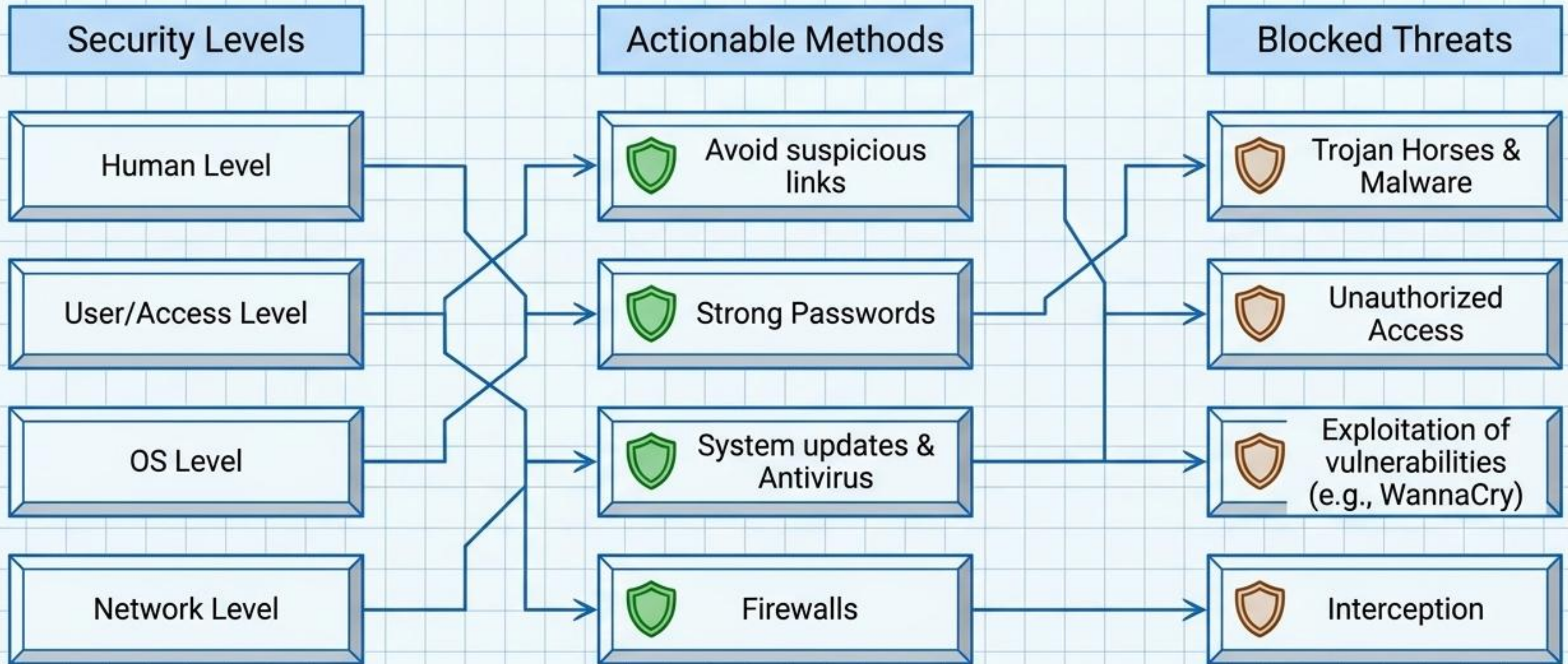
- **The Mechanism**

Uses encryption algorithms (like the Public Key) to create a digital signature that confirms the sender's identity.

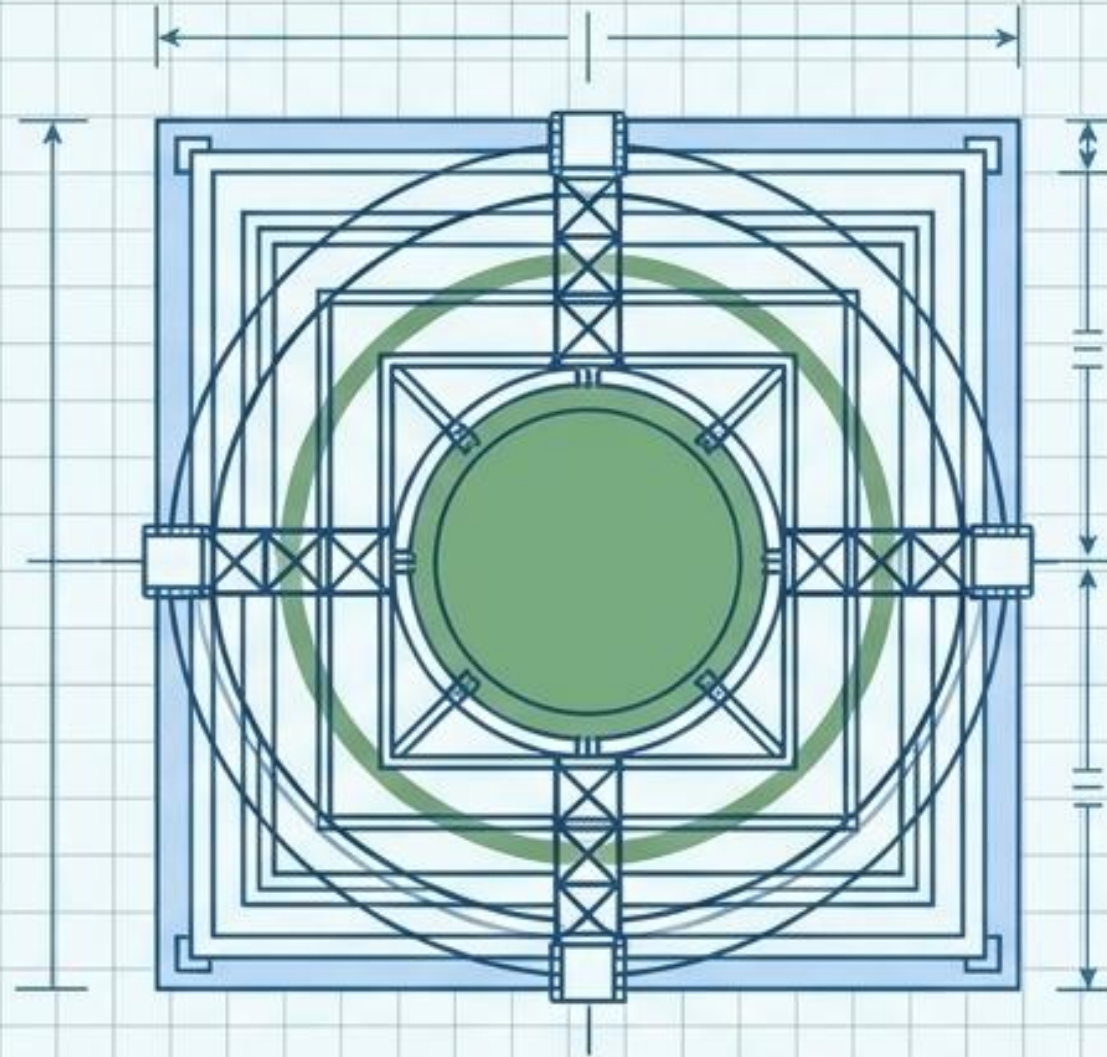
- **The Guarantee**

Ensures the message or document has not been tampered with or altered in transit, directly upholding the core security objective of Integrity.

Comprehensive Architecture Flow



Security is a Journey, Not a Destination



STATUS: Security is not a one-time setup; it is an essential, continuous operation that never stops.

STRATEGY: Systemic survival requires multiple, redundant layers of protection to ensure the entire system does not collapse if a single layer is breached.